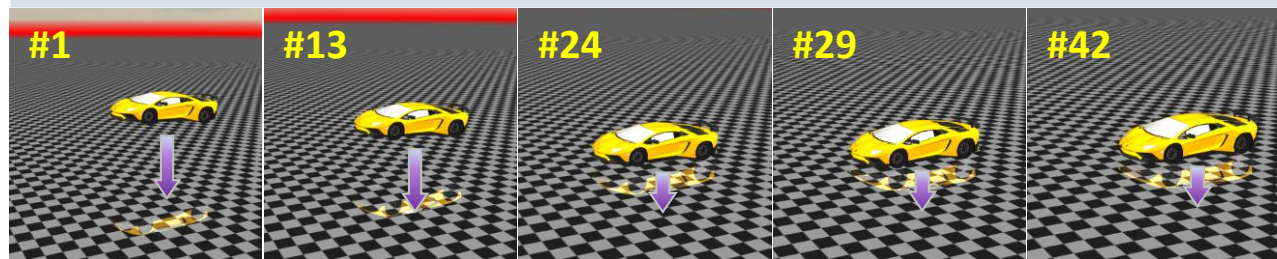
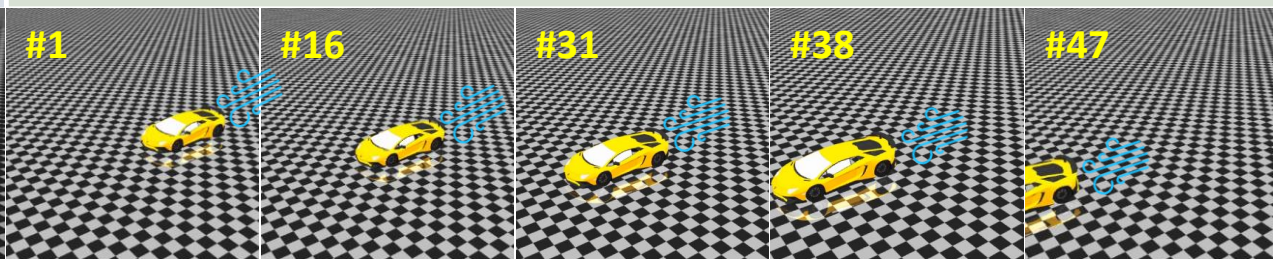


Free-Fall Motion Simulation



Wind-Driven Motion Simulation



Overall information

A1: Name: Lamborghini
 Category: Sports Car
 Dimension: 478*203*113
 Parts: l_0: Car Body, 1, Carbon Fiber Reinforced Polymer, 1.6 g/cm³, 70.0, 0.3, The main chassis and body shell of the Lamborghini...
 Group_info: group_0: [l_0', 'l_2', 'l_4', 'l_5', 'l_6', 'l_8', 'l_9', 'l_10', 'l_12']; Type: E;...

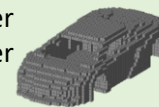
Sim-Ready Physical 3D Asset

XML

URDF

Geometry information

A2: a:\nb:1536 128\nc:1379 24;1443
 24;1507 3;1529 2;1536 128\n0:layer
 a\n1:layer a\n2:layer a\n3:layer
 a\n4:layer a\n5:layer a\n6:layer
 a\n7:layer a\n8:layer a\n9:
 layer b\n10:layer b\n11:...



Decoder
 Library

Part-level Meshes

Part-level RFs

Part-level 3DGS

Round 1

Vision-Language Model (VLM)



Image Tokenizer

Text Tokenizer



or

Full Image

Occluded Image
from scene

Image Prompt

Q1: Please analyze the given image of an object and output its structured description in the following format (voxel grid=64):
 <Overall format details>

Text Prompt



Vision-Language Model (VLM)



Image tokens Q1 tokens A1 tokens

Content memory



Text Tokenizer

Q2: Based on the structured description of l_0, generate its 3D voxel (grid=64) in the 3D RLE (linear scan) format. Output one run per line as: start_index length:
 <Geometry format details>

Text Prompt

Round 2 - Round n